CS4500 Responsible Computer Science
DSAIT4500 Responsible Data Science and AI Engineering

Group Work Report

Group X

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April 2026

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1 Context

In this report, we address the challenge posed by the discontinuation of the Dutch Crime Anticipation System (CAS) at the end of 2025. CAS predicted crime hotspots on a 125×125 meter grid heatmap based on historical crime data. Despite its ambitions, it was never satisfactorily evidenced to be effective [1], [2]. The Dutch Court of Audit (Algemene Rekenkamer) raised concerns about discriminatory bias and privacy risks, while research showed that human subjectivity inevitably shaped how officers interpreted CAS outputs [3].

We were asked: if the national police approached us for a better technical tool for crime prevention — one that is data-driven yet keeps humans skeptical of automated predictions and in control of final decisions — what would we propose?

The following chapters present our individual ideas (Section 2), subgroup proposals with interaction designs (Section 3), cross-audit feedback and revisions (Section 4), and our generative AI disclosure (Section 5).

2 Individual Thoughts and Questions

2.1 Orhan Ağaoğlu

2.1.1 Individual Proposal

When I first read about CAS, the thing that stuck out to me was that it was basically just showing where crimes happened before and assuming they'd happen there again. But that doesn't really tell you anything useful. If there's a lot of theft in some area, sure you can send more police there, but that doesn't solve why people are stealing in the first place.

What I'd want to build instead is something that looks at the reasons behind the crime. When you see crime going up somewhere, check what else is happening — are people losing jobs, is it hard to see a doctor, are kids dropping out of school, are utility bills going unpaid? Basically try to connect the crime to the conditions around it.

This would still be a police tool, but it changes what they actually do with the information. Instead of just patrolling an area more, the police would use the system to flag that something deeper is going on and pass it to whoever can actually help — the municipality for infrastructure, GGD for healthcare, social services for support programs. Police can't fix poverty or open a clinic, but they can be the ones to say "this area needs something." That's still crime prevention, just not the patrol-more-cars kind.

I'd also want different people to see different things. An urban planner just sees "this area needs attention." A health department gets the healthcare numbers. The police see that a referral was made but don't get the socioeconomic details — that's not their job. You only see what you can actually do something about. The system doesn't tell anyone what to do, it just goes "hey, look here." Then people figure it out.

2.1.2 Questions for Experts

Question 1 — To someone who actually used CAS in the Dutch police
Expertise: Day-to-day experience with CAS as a police analyst or officer.
Question: What was CAS actually doing for you on a normal workday? Did you find the heatmap helpful or did it mostly just sit there?
Why: I want to know what the real problem was before I try to fix it. What CAS was supposed to do and what it actually did are probably pretty different.

Question 2 — To a legal expert in Dutch law
Expertise: Dutch administrative and criminal law, especially around discrimination.
Question: If a municipality notices socioeconomic indicators going downhill in an area, are they allowed to put extra resources there just based on that? At what point does proactive policy become profiling? Galic, Das, and Schuilenburg (2023) raised similar legal concerns about Dutch predictive systems.
Why: My whole proposal is about acting before things get worse. But I'm not sure if acting on trends rather than actual problems crosses any legal lines.

Question 3 — To a sociologist who studies urban inequality
Expertise: How neighbourhoods get labelled and what happens to people when they do.
Question: If you start directing resources to an area, how do you avoid basically branding it as a problem area? Is there a way to help without making it worse through stigma?
Why: Even well-intentioned flagging could scare off businesses, raise insurance, or make people feel watched. I don't know how to avoid that.

2.2 Petar Verzaal

2.2.1 Individual Proposal

2.2.2 Questions for Experts

2.3 Alex Hautelman

2.3.1 Individual Proposal

Tool Proposal: Two-Tier Crime Prevention Support System

Proposal. I propose a two-tier decision-support system that helps police work more efficiently by combining geographic crime pattern mapping with a targeted monitoring component for people on supervised release. Crucially, the system does not make decisions; instead, it gives officers better information to act on.

Tier 1 — Crime Heatmap. A straightforward statistical model pulls together confirmed police reports and validated historical data to build a real-time map of where criminal activity is more likely to occur. This helps officers decide where to deploy faster and more deliberately. Public tip-offs can feed in too, but are down-weighted because some neighbourhoods get disproportionately reported regardless of what is actually happening there. The model is simple and explainable and handles only aggregated location data, so no personal information is involved. A built-in audit mechanism regularly checks whether the underlying data is skewing results due to historical over-policing of certain areas.

Tier 2 — Repeat Offender Monitoring. This component focuses on people with prior convictions under monitoring conditions set by a court, similar to how parole already works. The legal authority comes from the judge, not from asking people to consent (which would not be meaningful consent anyway). The focus is justified: 47% of adult ex-prisoners reoffend within two years of release [4], making this a sensible group to pay closer attention to. Monitoring is time-limited, restricted to a minimal and clearly defined set of signals, and feeds into the Tier 1 heatmap. When a geographic hotspot overlaps with a Tier 2 signal, that combined indicator gets weighted more heavily, giving officers a stronger, more specific prompt to act.

When humans and the model disagree. An officer's judgment always wins. The system has no authority to direct anyone. But disagreements should be logged, not ignored. If officers are consistently overriding the system in certain areas or situations, that is valuable feedback that can be used to retrain and improve the model over time. This keeps humans in control while still letting the system learn.

2.3.2 Questions for Experts

Question 1 — To a legal or privacy expert (GDPR / EU AI Act)

Question: What kinds of monitoring can actually be imposed as a court condition under Dutch and EU law, and what safeguards need to be in place to make sure it stays proportionate?

Why: The whole legitimacy of Tier 2 depends on this. If the court-ordered framing does not hold for certain types of data collection, the design has to change before any technical work starts. This is the first question that needs an answer.

Question 2 — To a criminologist or recidivism researcher

Question: Is prior imprisonment actually a reliable predictor of future offending once you account for things like poverty, mental health, or lack of support after release, and does that relation hold across different types of crime?

Why: The 47% reoffending statistic [4] gives us a reason to focus on this group, but a general

correlation is not the same as a validated risk model. We need to understand what is actually driving the pattern before we build a system around it; otherwise, we risk encoding existing inequalities rather than genuine risk.

Question 3 — To a frontline police officer or operations lead
Question: When would you trust a heatmap signal enough to act on it, and when would you ignore it? What would the tool need to show you for it to actually be useful rather than just another screen to click past?
Why: A tool that officers routinely override without thinking, or blindly follow without questioning, does not work. Real operational trust has to be designed in from the start, and officers know things about local context and day-to-day constraints that system designers simply do not.

Question 4 — To an ethicist or civil liberties researcher
Question: What would need to be built into this system — things like expiry dates, independent audits, and hard limits on scope — to make sure it does not quietly expand beyond what it was originally designed to do?
Why: Surveillance systems have a well-documented tendency to grow over time. Independent ethical oversight should not be something bolted on after criticism; it should be part of the architecture from the beginning.

2.4 Mart van der Lugt

2.4.1 Individual Proposal

The Crime Anticipation System (CAS) was based largely on historical police data. This means that initially, the historical patterns of policing and the inherent biases of the police force were encoded in the predictions. If an area was patrolled more, as it was deemed less safe by the police force, the chance of catching crime in that area would be higher, even if the actual crime rate was not necessarily higher.

Furthermore, this creates a positive feedback loop. Areas that were deemed high-risk would be patrolled more, more crime would be caught, and the model would intensify the high-risk prediction even more.

In a future model, the key improvement would be to separate the data source for the model from the patrol intensity. To do this, we can use other indicators of crime rate, like insurance reports, break-in attempts, and safety surveys, which have a lower bias than crimes caught by police. Crimes reported by police patrols could still be used as data but should be controlled using the intensity of patrols in a certain area.

The most important change would be in the output of the model. Instead of just reporting the risk areas, the model should also report the uncertainty related to that risk area. If the model reports increased police-reported crime in an area that is also patrolled more, the human operator should be able to see that. Likewise, the operator should also be able to see which areas have increased insurance-reported crime while police-reported crime has remained consistent. The operator is then able to take these factors into account, prevent positive feedback loops, and limit inherent biases.

2.4.2 Questions for Experts

Question 1 — To a criminologist familiar with the impact of patrols on crime rates
Question: The model assumes that a higher patrol rate results in a higher detected crime rate, as more of the crime happening in an area is caught. Presumably, this caused a positive feedback loop in the CAS model. What is the exact studied relationship between these factors?

Furthermore, does increasing patrol intensity eventually result in lower crime rates, or are there more effective policing measures that can be taken?

Question 2 — To a statistician at the CBS
Question: An important input for the new model is safety surveys. Could you give an indication of the reliability of these surveys? Are there any important biases that should be considered? Furthermore, at what level do these surveys operate? Is it possible to get an indication of safety or crime rate at the street level, or at the neighbourhood level?

Question 3 — To a lawyer familiar with insurance and data protection
Question: The most important input for the model is crime information that is not necessarily reported by the police. The most straightforward data source would be insurance reports. Is this something that the police could get access to at a large scale? What privacy considerations should be taken? How could we convince the insurance companies to collaborate on this issue?

3 Subgroup Proposals

3.1 Subgroup A — Orhan Ağaoğlu & Petar Verzaal

3.1.1 Tool / Intervention Proposal

We're proposing something we call the **Community Structural Risk Monitor (CSRM)**. The idea is to give the police a way to do crime prevention that goes beyond just showing up more. Instead of a heatmap that says "crime happens here," CSRM tries to figure out *why* it's happening and helps the police coordinate with the right people to deal with it — municipalities, health services, social workers, housing corporations. The police are still running the show, they're just doing something smarter with the information than sending more patrols.

How it works: the system watches two things at the same time. One is regular crime reports from the police — stuff that already happened. The other is what we call stress signals — things like emergency calls, social services referrals, housing complaints, kids dropping out of school. These tend to show up before crime gets worse. When both of these start going up in the same area at the same time, the system opens up a 12-month window and starts pulling in neighbourhood-level data for that area.

The data it collects comes from places like CBS for income statistics, GGD for healthcare access, housing corporations for eviction numbers, and municipal records for things like benefit uptake. None of this is personal data — it's all aggregated at the neighbourhood level. That's a big difference from existing Dutch systems like Top600 in Amsterdam or ProKid 12-SI, which actually create risk scores for individual people [1]. We don't think that's the right approach. Our system looks at the area as a whole and checks if the conditions there match the crime trends. It never singles anyone out.

We also don't just throw in any data we can find. The indicators have to be validated by people who actually know the field — criminologists, sociologists, public health researchers — as stuff that's known to relate to crime. And the most important rule we came up with: you can only see the data if you can actually do something about it. Doctor shortages go to the health ministry. Unemployment numbers go to the labour office. Housing problems go to housing corporations. If you can't fix it, you can't see it. We've been calling this "access follows accountability."

The output is simple. A pin on the map, and only when everything lines up — crime going up, stress signals going up, and the structural indicators matching. No scores, no rankings, no red zones. Just a pin that says "something seems to be going on here, take a look." What happens after that is a human decision.

3.1.2 Human–Machine Interaction Design

The whole point of how CSRM works is that the system helps you notice things, it doesn't decide things for you. That sounds like it should be obvious but CAS showed how quickly it falls apart. Galic, Das and Schuilenburg (2023) wrote about how officers ended up treating the CAS heatmap as if it was telling them what to do, even though it was just a prediction [1], [3]. The map looked authoritative, so people treated it that way. We wanted to avoid that from the start.

3.1.2.1 How the Interaction Works

There are four stages to how someone uses CSRM:

1. **Nothing happens (most of the time).** The system is running in the background, watching the two signal streams across all neighbourhoods. Unless things actually start lining up, it stays quiet.
2. **A pattern shows up.** If the system picks up something — crimes happening in the same spot, same kind of crime, or similar situations for the people involved — it opens a data window and starts gathering the structural indicators for that area.
3. **A pin appears.** If the structural data actually matches the crime and stress pattern over the window period, a pin shows up on the map. That's the only thing the system ever does.
4. **People talk about it.** The pin goes out as a notification to the relevant departments. A panel meets, discusses whether something needs to happen, and decides. The system doesn't get a vote.

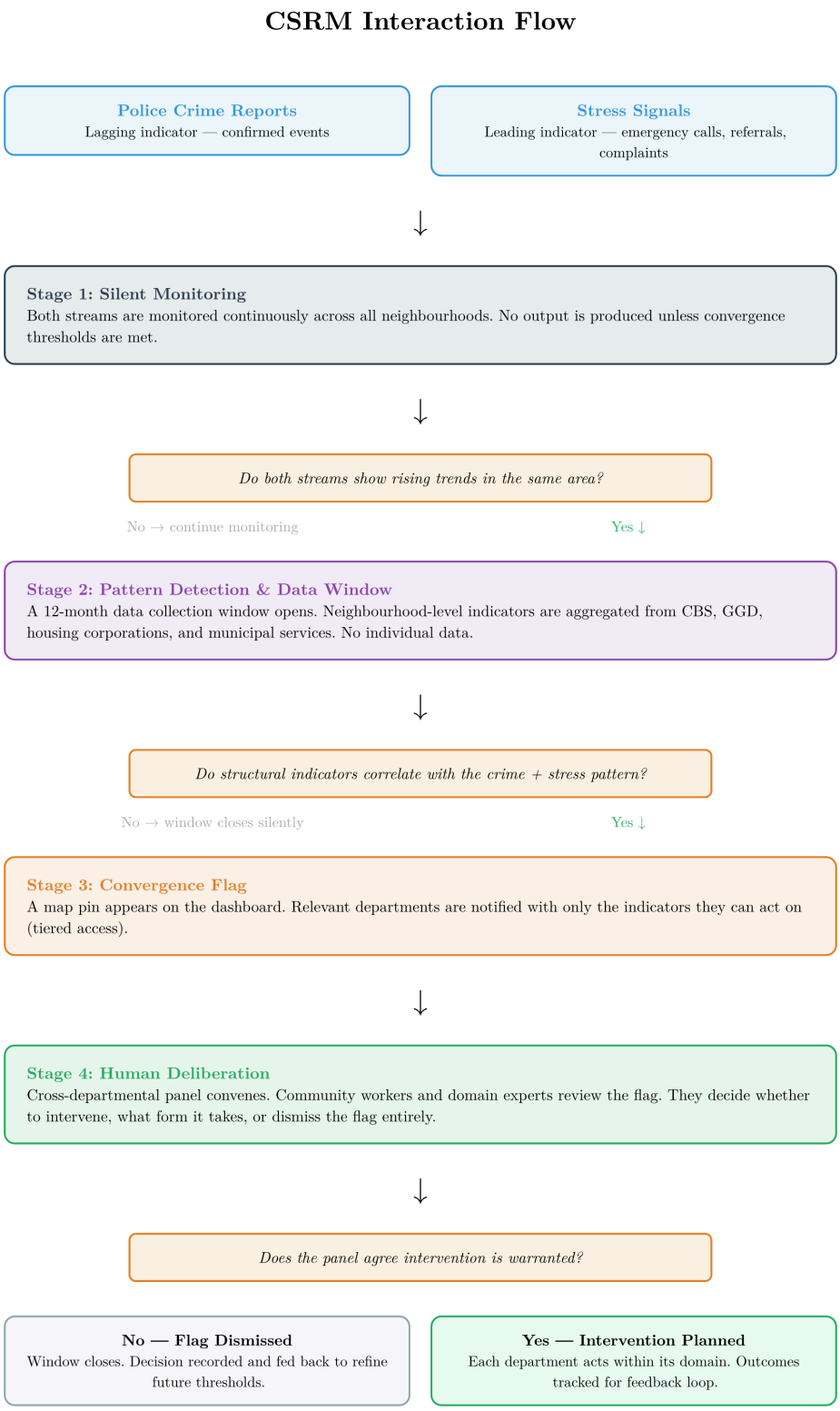


Figure 1: CSRM interaction flow: from monitoring to human deliberation.

3.1.2.2 When People Disagree with the System

We actually expect people to disagree with the flags, and that's fine. Pozzi, Sand, and Jongsma (2026) make the argument for medicine that disagreement between humans and AI isn't a problem to solve, it's actually useful [5]. Same idea here.

A community worker who's been in the neighbourhood for ten years knows stuff that no dataset is going to capture — who actually talks to who, what programs are already running, whether things are actually as bad as the numbers make them look. If they look at a flag and go "no, that's not what's happening," they should be the ones we listen to. The pin is basically asking "did you know about this?" not saying "do something about this."

That's the big difference from CAS. The CAS heatmap felt like a fact [3]. It had colours and squares and it looked like it knew something. Our system is just a dot on a map with a question behind it. It's saying "we noticed something, do you see it too?"

If the experts look at it and decide it's nothing, the window closes and that's the end of it. But that decision gets saved so the validation panel can use it to adjust the thresholds later. So over time the system gets better at knowing when to raise a flag because it's learning from the times humans said "no, you're wrong."

3.1.2.3 Who Sees What

Nobody gets to see everything. Each person only sees what's related to their job:

- **Police** see: that a flag was raised, who got notified, and whether anything came of it. They don't see the income data or the healthcare numbers — that's not what they're there for.
- **Urban planners** see: "this area needs attention." That's it, no details.
- **GGD / health services** see: healthcare stuff only — how long people wait for a GP, whether there's enough addiction support.
- **Labour department** sees: jobs and income data.
- **Housing corporations** see: eviction rates, how much rent is unpaid, building conditions.

This isn't just about privacy for the sake of it. There's a real problem in the Netherlands where data that was collected for one thing ends up being used for something completely different [1], [6]. If the police can't see the socioeconomic data, they can't use it to profile areas. If journalists can't get the flags, there's no "top 10 worst neighbourhoods" list. The way the system is built makes misuse hard, instead of just having a rule that says "please don't misuse this."

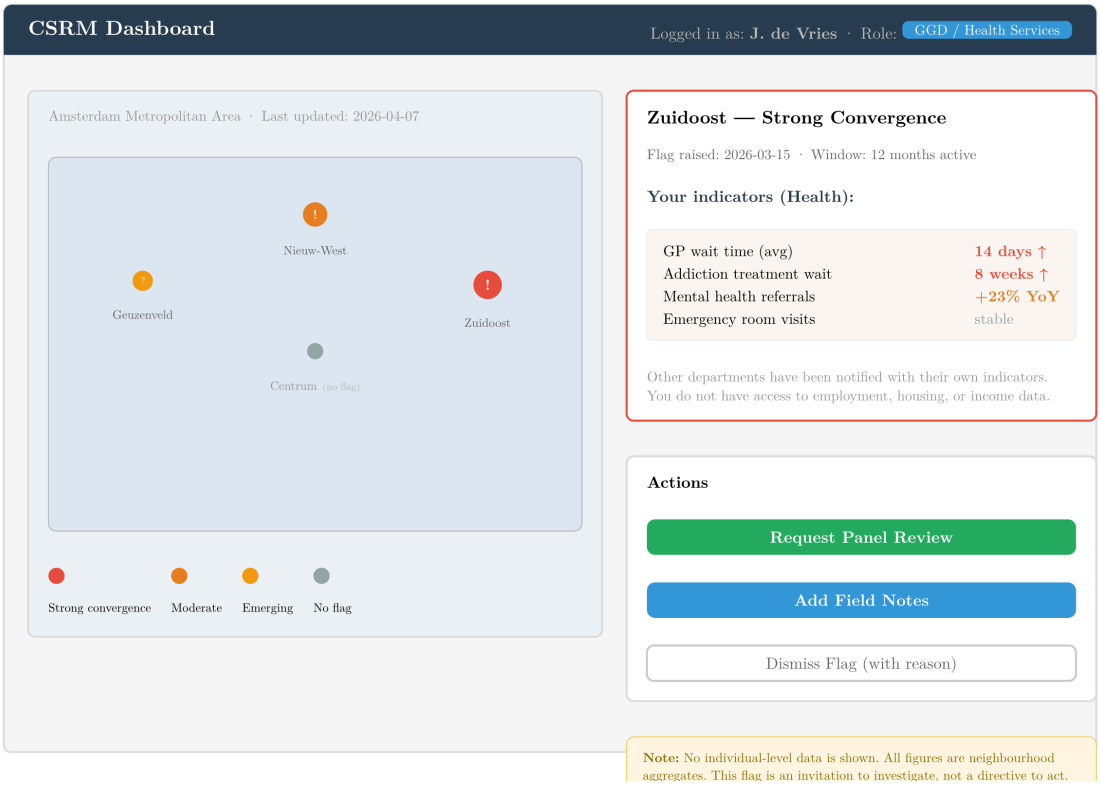


Figure 2: CSRM dashboard mockup — GGD health services view. The logged-in user only sees health-related indicators. Employment, housing, and income data are hidden because that's not their department.

3.2 Subgroup B — Alex Hautelman & Mart van der Lugt

3.2.1 Tool / Intervention Proposal

We propose the **Spatial Risk and Resource Allocation Dashboard (SRRAD)**: a decision-support tool designed to help police commanders allocate patrol resources more effectively by surfacing where and when crime risk is elevated. The tool divides a municipality into a fine-grained grid and assigns each cell a continuously updated risk score, visualized as an interactive heatmap. SRRAD is an advisory instrument only: it has no authority to dispatch officers, trigger alerts, or take any action. Every operational decision remains with the human commander. SRRAD deliberately avoids demographic data and raw arrest statistics, since these inputs can encode historical policing bias back into future predictions. Instead, the model draws exclusively on direct environmental and behavioral signals:

- Municipal infrastructure telemetry: broken streetlights, abandoned vehicles, recurring vandalism reports.
- Anonymized emergency service logs: police interventions with confirmed outcomes, ambulance dispatches for physical trauma, and firefighter callouts.
- Commercial security incidents: shoplifting, aggressive behavior, and trespassing reports from local businesses.
- Insurance claims: aggregated theft, robbery, vandalism, and arson data.
- Public tips: anonymized suspicious-activity reports from the public tip line.
- Safety surveys: community-level perception data.

- Live deployment data: police presence per grid cell to contextualize scores and detect feedback-loop inflation.
- Electronic monitoring signals: anonymized activity flags from individuals under judicial supervision.
- Prediction feedback: structured post-patrol debriefs allowing officers to confirm, challenge, or correct prior assessments.

Alongside each risk score, SRRAD displays a confidence interval and a bias risk flag. Built on an interpretable statistical framework like generalized additive models, every score can be traced back to its specific underlying signals.

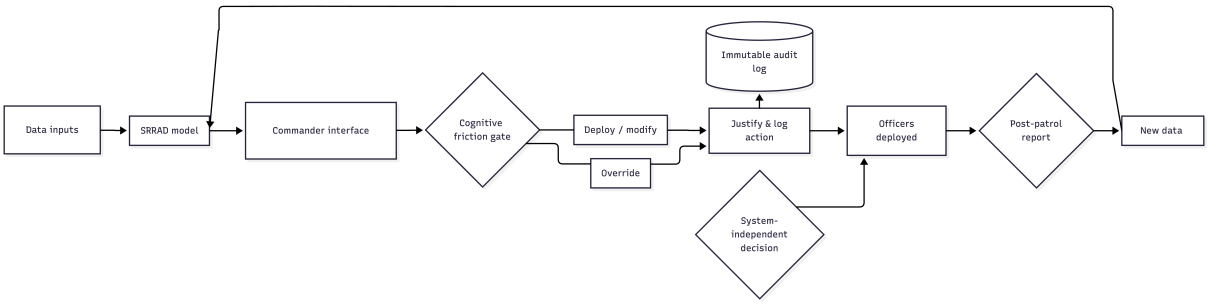
3.2.2 Human–Machine Interaction Design

A major risk in any decision-support system is that users stop questioning it. Under pressure, people naturally defer to a confident-looking number on a screen. In predictive policing, this tendency is dangerous. It can concentrate police presence in already over-policed areas, erode officer judgment, and create legal and ethical exposure.

SRRAD uses a human-in-command governing principle. The interface is actively designed to slow down deference, force engagement with the underlying data, and keep responsibility strictly with the officer making the call.

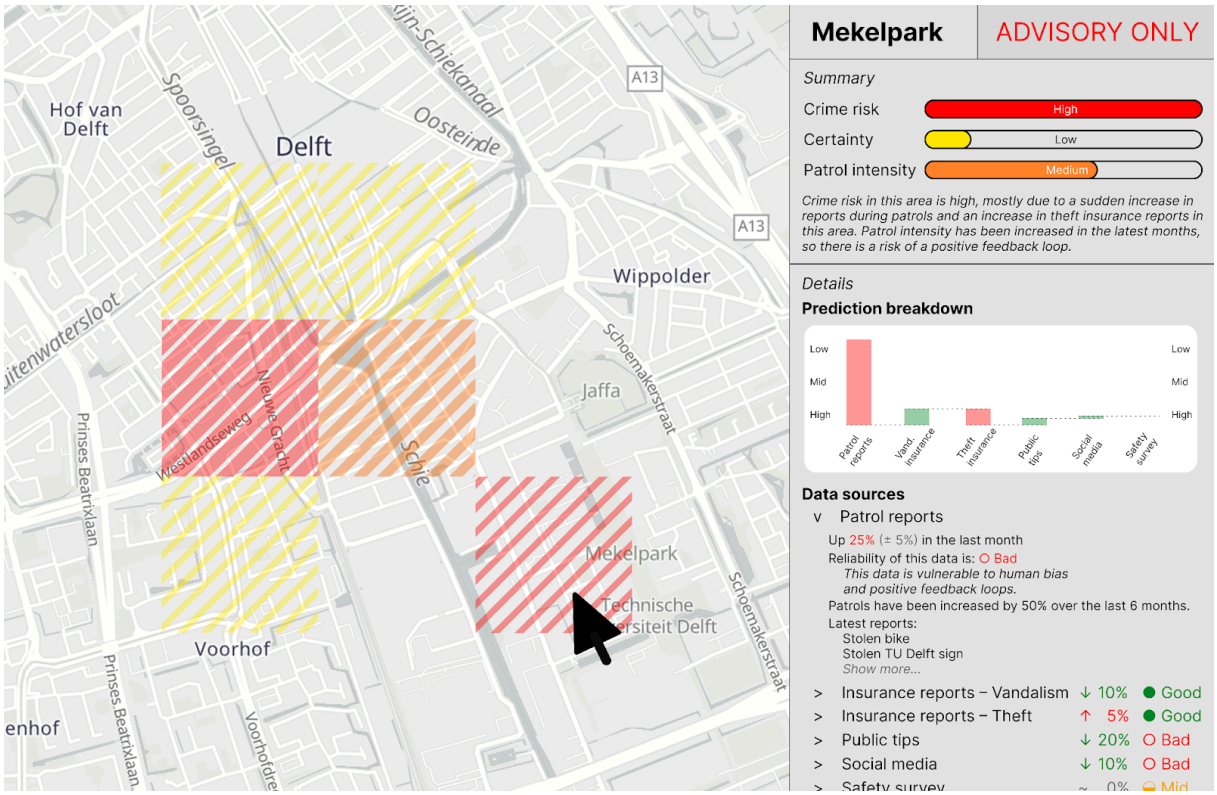
3.2.2.1 The Interaction Flow

The diagram below illustrates how information moves from raw data sources through to a logged deployment decision and back into the model.



3.2.2.2 Page 1: Risk Heatmap

The main screen shows the grid overlaid on a municipal map. Each cell is color-coded by risk score, but the display deliberately foregrounds uncertainty. Cells with low data coverage show a reduced-opacity fill, and cells driven by a single data source carry a visible bias risk flag. A commander scanning the map immediately sees how confident the model is and whether that confidence is warranted. The “Advisory Only” banner is permanent, serving as a legal reminder that the heatmap is a starting point for judgment, not a conclusion.



3.2.2.3 Page 2: Prediction Breakdown and Command Decision Log

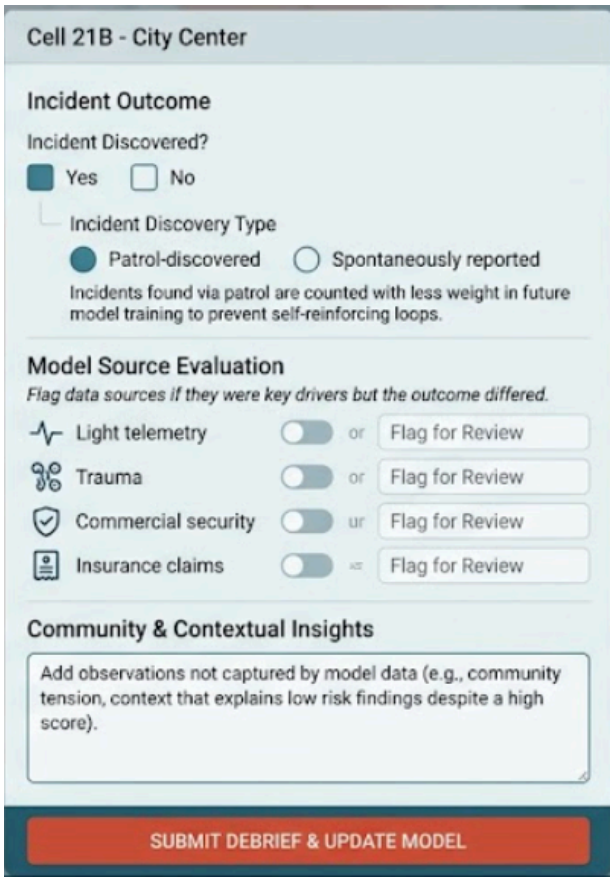
Clicking a cell opens the prediction breakdown, which translates feature importance into plain-language drivers. Seeing that a score is driven by broken streetlights might prompt a commander to request municipal repair services instead of deploying a patrol unit.

Crucially, there is no quick-deploy button. To log any action, the commander must acknowledge the confidence level, select an action type, and enter a brief written justification. This creates a cognitive friction gate, a deliberate pause that reduces blind deference. It also produces a timestamped, immutable log linking each decision to the model version and the stated reasoning.

3.2.2.4 Page 3: Prediction Correction and Post-Patrol Debrief

The third screen manages the feedback loop and protects the long-term integrity of the model. After a patrol, officers complete a structured debrief. First, they log if an incident was found and whether it was patrol-discovered or spontaneously reported. Patrol-discovered incidents get a discounted weight in future training to prevent a self-reinforcing loop where police presence inflates future scores.

Second, the screen allows officers to flag data sources they believe are improperly weighted. Finally, it captures qualitative observations the model cannot see, such as community tension or visible disorder, which are attached to the audit log for analysts.



3.2.2.5 Addressing the Challenges

This design directly responds to the three tensions identified in the stakeholder analysis. Automation bias is countered by the cognitive friction gate and the requirement to articulate a justification. The feedback-loop problem is addressed structurally by discounting patrol-discovered incidents during model training. Finally, accountability and liability are made fully traceable through the immutable audit log, making it clear that operational responsibility rests with the commander who made the decision.

4 Cross-Audit: Subgroup A's Proposal

Strengths

The proportionality logic is solid: data access only widens when multiple independent signals converge, and "access follows accountability" is baked into the architecture rather than left as a policy promise. Treating the output as a question rather than a verdict is a meaningful guard against over-reliance, and building in a feedback loop where expert disagreements improve future thresholds is the kind of human-in-the-loop design that makes systems like this more accurate and trustworthy over time. The cross-departmental panel also sets a healthy bar for turning a flag into action.

Areas for Improvement and Open Questions

Signal completeness and structural bias. The system may be quietest in the places that need it most. Housing signals are easier to generate in rental stock than in owner-occupied homes; school dropout signals depend on schools having social workers to notice them, which is less likely in under-resourced schools. The result is that a well-served neighbourhood having a rough patch could trigger the system faster than a chronically stressed one where data barely gets recorded. One practical fix: weight signals by source coverage. A referral from a poorly covered area is arguably stronger evidence than one from a well-monitored one, because it is harder to produce.

Convergence criterion and baseline blindspots. If both crime and stress signals need to go up before a flag is raised, areas where stress has been high for years may never trigger since there is no upward movement to detect. Chronic deprivation is likely more criminogenic than a sudden short-term spike, so the convergence requirement may miss the neighbourhoods most at risk. It is worth considering whether hitting an absolute threshold, and not just a relative increase, should also be enough to highlight the existing stress. Is the stress-crime relationship modeled as linear, and if not, how does the system capture that?

Historical bias in thresholds. Calibrating on historical data means inheriting the biases of whoever was recording crime and distributing services at the time. A practical safeguard would be periodic bias audits, comparing where the system flags against independent assessments of neighbourhood need, with results shared with the validation panel. Who is currently responsible for noticing when the model appears systematically off, and what is the process for correcting it?

The 12-month window — historical or prospective? It is unclear whether the system looks back at existing data or starts tracking from the moment a flag is raised. These serve different purposes: historical data helps explain how a situation developed, live tracking shows whether it is getting better or worse. Combining both, showing historical context alongside current indicators in the panel view, could make the flag much more useful to the people acting on it.

Collaboration under siloed access. When a flag is raised, a labour department analyst and a GGD researcher looking at the same area may each have part of the explanation but no way to compare notes. More broadly, insights derived from publicly funded data sources — CBS, GGD, municipal records — arguably should not be locked behind departmental access controls at all. Making aggregate structural findings public by default, while keeping operational flags restricted, would naturally enable cross-departmental collaboration without requiring new infrastructure, and would increase transparency as a side effect. What is the reasoning behind keeping these insights internal, and has a public-by-default model for outputs been considered?

Bad-faith scenarios. A flag existing at all is sensitive, even if the underlying data is hidden. Police who know an area has been flagged may act on that alone, which is exactly the automation bias the design is trying to prevent. Framing notifications as "a panel has been convened for area X" rather than "area X has been flagged" is a small change that better reflects the deliberative intent. On the access side: what stops an actor like a housing corporation from using their partial data view in ways that serve their own interests? Access logging with anomaly detection reviewed by an independent auditor would be a reasonable addition.

4.1 Response by Subgroup A

4.2 Revised Proposal by Subgroup A

4.2.1 Revised Tool / Intervention Proposal

4.2.2 Revised Human–Machine Interaction Design

5 Cross-Audit: Subgroup B's Proposal

5.1 Critical Audit by Subgroup A

There is a lot to like about SRRAD. The way the proposal identifies how people behave under decision-making pressure and then designs around that is strong. The reasoning from "officers will defer to a confident-looking number" to "so we build friction into the interface" makes sense, and the cognitive friction gate — where you have to acknowledge the confidence level, pick an action, and write a justification before anything happens — is a genuinely good idea. It slows things down in a way that forces the officer to actually think instead of just clicking through. That is more humane, not necessarily better, but it puts a human back in a process that easily becomes automatic.

That said, we have some concerns.

The Bias Risk Flag is contradictory. SRRAD shows a confidence interval and a bias risk flag alongside each score. But if the system is telling you "this prediction might be biased," what is the officer supposed to do with that? They are not a data scientist. And more fundamentally, you are asking the same system to judge its own reliability. That is circular. If we do not trust the model enough to use it without a bias warning, maybe the question is whether we should be showing that score at all, not whether we should put a small flag next to it.

Grid-based districts create artificial boundaries. Dividing a city into a grid and coloring cells by risk level means you are drawing lines that do not exist in reality. One side of a street is "high risk," the other is "low risk," based on where an arbitrary boundary fell. That worries us. It is the same kind of labeling problem that heatmaps have always had — you are attaching a risk score to a place, and everyone who lives or works in that place inherits that label whether they had anything to do with it or not. We would like to hear how SRRAD deals with edge effects and whether the grid resolution was chosen for a reason beyond technical convenience.

Environmental signals can be proxies for poverty. The proposal says it avoids demographic data to prevent bias, which is good. But inputs like broken streetlights, abandoned vehicles, and recurring vandalism reports are strongly correlated with municipal underfunding and poverty. A neighbourhood with an underfunded council will always have more broken infrastructure, and under this model, it will always score higher on risk. So the system might end up measuring deprivation rather than crime risk, which brings the bias in through a different door. How do you separate genuine crime signals from signals that just reflect that an area is poor?

The feedback loop might not actually be solved. Discounting patrol-discovered incidents during model retraining is a smart idea. But the commander still sees the heatmap, still sends patrols to the red cells, and those patrols still generate reports from those areas — even if the reports are weighted less. Over time, does the discount actually break the loop, or does it just slow it down? We would want to see evidence or at least an argument for why the discount rate is enough.

Adversarial users are not addressed. The friction gate assumes good faith. A commander writes a justification, and that goes into the audit log. But what stops a bad-faith officer from typing "high risk score, deploying as recommended" every single time? The log would look compliant, but no real human judgment happened. Every IT system has adversarial users, and in a policing context the consequences of gaming the system are serious — you could use SRRAD to justify patrols you already wanted to do in certain neighbourhoods. The proposal

needs to address what happens when someone actively misuses the tool, not just when someone passively over-relies on it.

The flow diagram could go further. The current diagram shows the path from data to decision, but the steps between a commander seeing the heatmap and actually filing a report or making a deployment could be more detailed. What happens if two commanders disagree about the same cell? What if the debrief contradicts the original score — who resolves that, and how quickly does it feed back?

5.2 Response by Subgroup B

We thank Subgroup B for their rigorous audit of SRRAD. Their analysis highlights valid concerns regarding our spatial design, environmental data proxies, and human-machine interaction framework. We recognise that features intended for transparency (such as the bias flag and friction gate) could cause confusion or enable adversarial misuse. Below are the concrete improvements we will integrate into the revised proposal.

1. Redefining Metrics and Data Integrity

The audit noted that presenting both a confidence interval and a bias risk flag shifts an unfair analytical burden onto officers and creates circular logic. Furthermore, discounting patrol-discovered incidents may only slow, rather than break, the feedback loop.

- **Unified Certainty Metric:** We will replace the independent bias flag with a single "Certainty" metric. Certainty will only increase when multiple, heavily weighted, and independent signals corroborate a prediction.
- **Upstream Bias Mitigation:** Data engineers, not officers, will handle bias. Biased sources will be aggressively filtered, unbiased sources will be handled during preprocessing, or biased sources will be excluded from the model entirely.
- **Feedback Loop Thresholds:** We will introduce a saturation threshold. If an area receives repeated deployment recommendations, the system will pause further scoring until data analysts conduct a mandatory manual recalibration.

2. Spatial Representation and Environmental Proxies

We agree that arbitrary grid lines create artificial edge effects and that infrastructure telemetry (e.g., broken streetlights) risks measuring poverty rather than crime, potentially reinforcing institutional prejudice.

- **Contextual Zoning:** We will eliminate the arbitrary grid. SRRAD will instead use amorphous, dynamically generated heat zones or rely strictly on established municipal and socio-economic boundaries.
- **Expert Validation:** We will consult criminologists to rigorously separate genuine crime-driving signals from baseline poverty indicators before weighting telemetry data.
- **Cross-Agency Interventions:** We will leverage environmental data to shift responses away from purely punitive patrols. When risk scores are driven by infrastructure decay, SRRAD will explicitly prompt commanders to request municipal interventions (e.g., street repair) instead of police deployment.

3. Human-Machine Interaction and Governance

The audit rightly pointed out that our cognitive friction gate assumes good-faith users and fails to account for adversarial "rubber-stamping." The initial flow diagram also lacked necessary operational depth.

- **Multi-Officer Authorisation:** For deployments in previously unpatrolled or highly sensitive areas, SRRAD will require consensus. Two commanders must independently review the data and log a joint decision.
- **Audit Log Peer Review:** We will introduce mandatory, random administrative reviews of command decision logs. Officers identified as habitually bypassing the friction gate with generic, bad-faith entries will face operational penalties.
- **Expanded Operational Flow:** We will update the flow diagram to map edge cases. This will detail pathways for resolving commander disagreements, managing contradictions between the initial score and post-patrol debriefs, and establishing strict timelines for how quickly feedback updates the live model.

5.3 Revised Proposal by Subgroup B

5.3.1 Revised Tool / Intervention Proposal

We propose the **Spatial Risk and Resource Allocation Dashboard (SRRAD)**: a decision-support tool designed to help police commanders allocate patrol resources more effectively by surfacing where and when crime risk is elevated. The tool divides a municipality into [Updated] contextual zones based on established municipal boundaries and socio-economic similarities, replacing arbitrary grids to prevent edge-effect labeling and artificial division. SRRAD assigns each zone a continuously updated risk score, visualised as an interactive heatmap.

SRRAD is an advisory instrument only: it has no authority to dispatch officers, trigger alerts, or take any action. Every operational decision remains with the human commander. SRRAD deliberately avoids demographic data and raw arrest statistics, since these inputs can encode historical policing bias back into future predictions. Instead, the model draws exclusively on direct environmental and behavioral signals:

- [Updated] Municipal infrastructure telemetry (expert-validated to separate genuine crime signals from baseline poverty indicators): broken streetlights, abandoned vehicles, recurring vandalism reports.
- Anonymized emergency service logs: police interventions with confirmed outcomes, ambulance dispatches for physical trauma, and firefighter callouts.
- Commercial security incidents: shoplifting, aggressive behavior, and trespassing reports from local businesses.
- Insurance claims: aggregated theft, robbery, vandalism, and arson data.
- Public tips: anonymized suspicious-activity reports from the public tip line.
- Safety surveys: community-level perception data.
- Live deployment data: police presence per grid cell to contextualize scores and detect feedback-loop inflation.
- Electronic monitoring signals: anonymized activity flags from individuals under judicial supervision.
- Prediction feedback: structured post-patrol debriefs allowing officers to confirm, challenge, or correct prior assessments.

[Updated] Alongside each risk score, SRRAD displays a single, unified "Certainty" metric. We have removed independent bias risk flags to avoid placing circular analytical burdens on officers. Instead, bias is aggressively filtered and un-biased upstream by data engineers. The Certainty metric only increases when multiple, heavily weighted, and independent signals corroborate a prediction.

5.3.2 Revised Human–Machine Interaction Design

Human–Machine Interaction Design

A major risk in any decision-support system is that users stop questioning it. Under pressure, people naturally defer to a confident-looking number on a screen. In predictive policing, this tendency is dangerous. It can concentrate police presence in already over-policed areas, erode officer judgment, and create legal and ethical exposure.

SRRAD uses a human-in-command governing principle. The interface is actively designed to slow down deference, force engagement with the underlying data, and keep responsibility strictly with the officer making the call.

The Interaction Flow

The diagram below illustrates how information moves from raw data sources through to a logged deployment decision and back into the model.

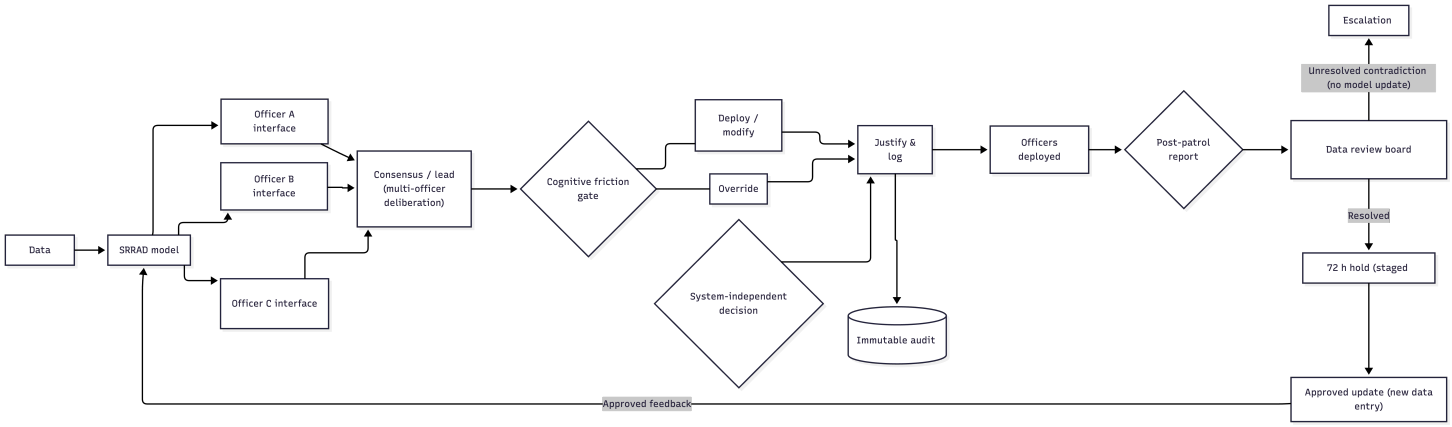


Figure 6: The flow diagram has been expanded to map edge cases. It now includes detailed pathways for multi-officer consensus loops, managing contradictions between the initial score and the debrief, and strict temporal rules for how quickly feedback updates the live model.

Page 1: Risk Heatmap

The main screen shows the [Updated] amorphous, dynamically generated heat zones overlaid on a municipal map. Each [Updated] zone is color-coded by risk score, but the display deliberately foregrounds uncertainty. Zones with low data coverage show a reduced-opacity fill. A commander scanning the map immediately sees how confident the model is based on the [Updated] Certainty metric. The "Advisory Only" banner is permanent, serving as a legal reminder that the heatmap is a starting point for judgment, not a conclusion.

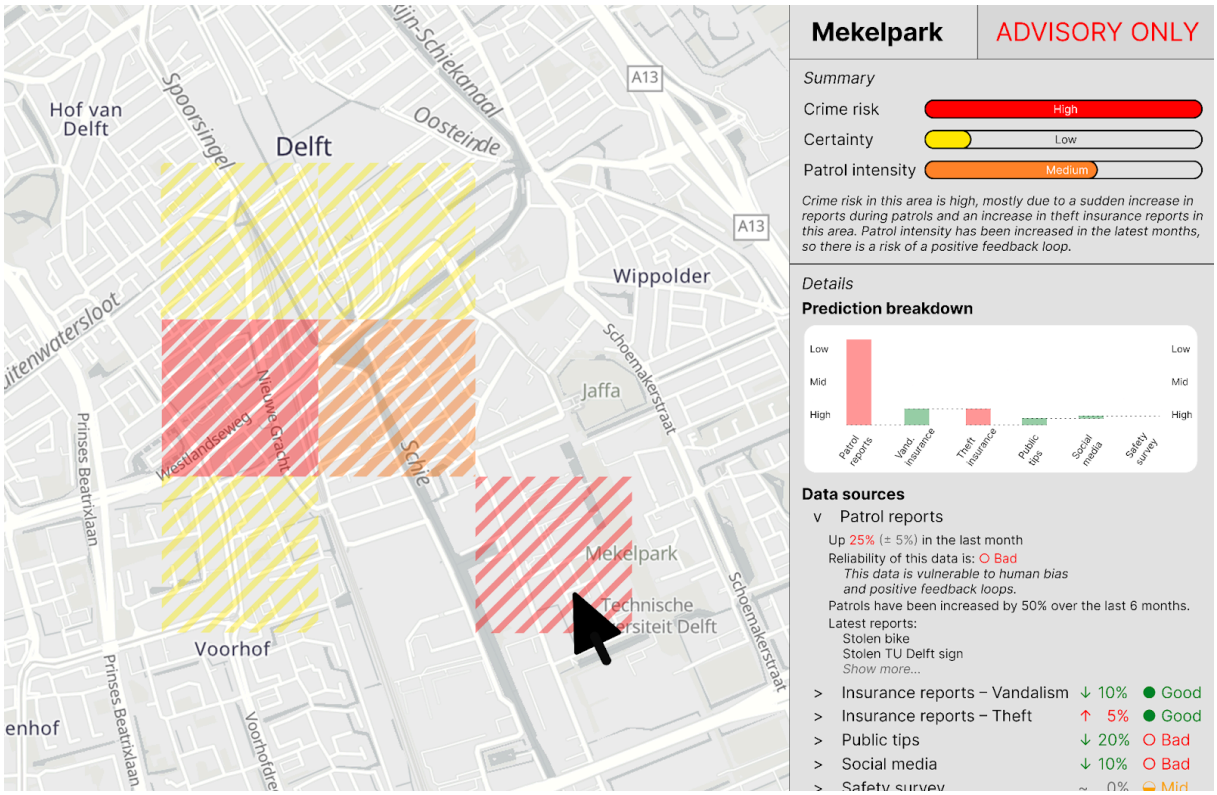


Figure 7: Remove arbitrary grid lines in favor of amorphous zones or municipal boundaries.

Page 2: Prediction Breakdown and Command Decision Log

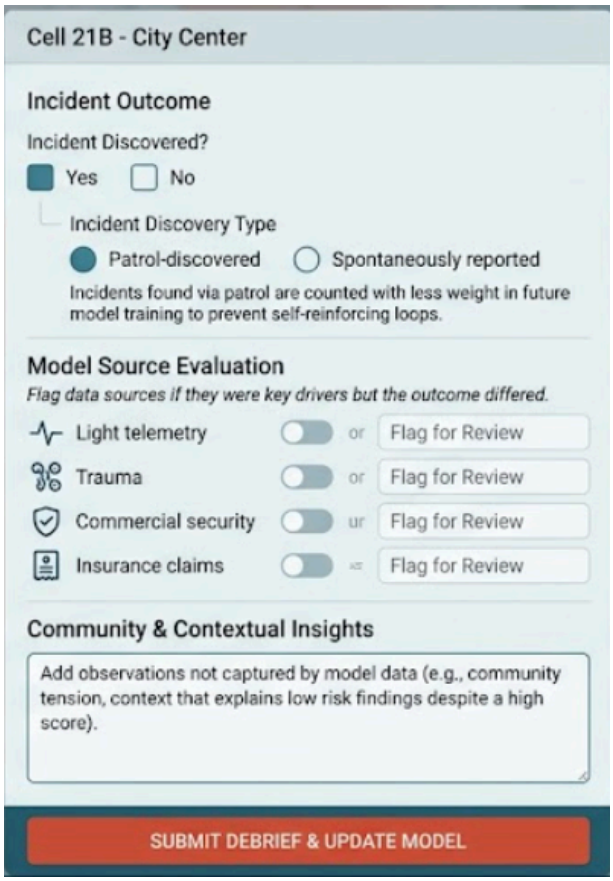
Clicking a cell opens the prediction breakdown, which translates feature importance into plain-language drivers. [Updated] We emphasize cross-agency interventions: seeing that a score is heavily driven by infrastructure decay (e.g., broken streetlights), SRRAD explicitly prompts the commander to request municipal repair services instead of deploying a punitive patrol unit.

Crucially, there is no quick-deploy button. To log any action, the commander must acknowledge the Certainty level, select an action type, and enter a brief written justification. This creates a cognitive friction gate, a deliberate pause that reduces blind deference. [Updated] To counter adversarial users who might approve decisions without consideration, deployments in previously unpatrolled or highly sensitive areas require multi-officer authorisation: two commanders must independently review the data and log a joint decision. Furthermore, all logs are subject to mandatory, random administrative peer reviews.

Page 3: Prediction Correction and Post-Patrol Debrief

The third screen manages the feedback loop and protects the long-term integrity of the model. After a patrol, officers complete a structured debrief. First, they log whether an incident was found and whether it was patrol-discovered or spontaneously reported. Patrol-discovered incidents get a discounted weight in future training. [Updated] To guarantee this discount breaks the feedback loop, we introduce a saturation threshold: if an area receives repeated deployment recommendations, the system pauses further scoring until data analysts conduct a mandatory manual recalibration.

Second, the screen allows officers to flag data sources they believe are improperly weighted. Finally, it captures qualitative observations the model cannot see, such as community tension or visible disorder, which are attached to the audit log for analysts.



Addressing the Challenges

This design directly responds to the tensions identified in the stakeholder analysis. Automation bias is countered by the cognitive friction gate and the requirement to articulate a justification. [Updated] Adversarial misuse is mitigated through multi-officer consensus rules and mandatory peer reviews of audit logs. The feedback-loop problem is addressed structurally by discounting patrol-discovered incidents [Updated] and enforcing saturation thresholds that mandate human recalibration. Finally, accountability and liability are made fully traceable through the immutable audit log, making it clear that operational responsibility rests with the commander who made the decision.

6 Generative AI Disclosure

We used AI during this assignment and are disclosing it here as required.

6.1 Tools and Services Used

- **Claude Code (Anthropic)** — used through the Claude Code CLI as an research assistant and visual designer. Addionaly for organization and documentation. (We used github for version control and Typst for writing, claude was used during the comands and edits of these tools)

6.2 Contributions by Section

6.2.1 Section 2: Individual Thoughts and Questions

Orhan Ağaoğlu: I used Claude as a librarian while developing my proposal. The ideas themselves (looking at causes instead of crime locations, making the police a coordinator rather than the end user, tiered access) came from me. Claude helped me understand some of the terminology — things like ecological correlation and data minimisation — and explained what they actually mean so I could decide if they fit my proposal. I also asked Claude to help me understand the Galic et al. (2023) paper on CAS and Dutch predictive systems, since it's referenced in the assignment and I wanted to know what it actually says before citing it. I had initially a completely diffent idea but existing systems in Dutch police had me change my thoughts.

6.2.2 Section 3: Subgroup Proposals

The technical design came out of the conversation, with me describing how I thought things should work and Claude helping put it into words you see in the diagram. Claude also generated the flow diagram and the dashboard mockup — I described what I wanted them to show and Claude built them in Typst. I also used Claude to look up and understand specific terms and concepts that came up during the design (e.g., what Top600 and ProKid 12-SI actually do, how ecological correlation works in practice, what the Dutch Court of Audit report said about CAS).

Prompts used: "create a mockup and flow diagram based on my proposal." (plus additional manipulation on Typst)

6.3 Motivation and Possible Impacts

We used AI to speed up things we'd have spent a long time on otherwise — reading through papers, figuring out what specific terms mean, and making diagrams. The design decisions and arguments are ours. One downside we ran into is that AI output is usually narrower than what we are trying to achieve in this course. If we hadn't used AI only as an assisntant for our clear purposes and aligned perspective, the report would've sounded like it was written by a different person entirely.

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